

KARAN SINGH RATHORE

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PROFESSIONAL SUMMARY

Third-year B.Tech CSE (AI/ML) student with hands-on experience building and deploying machine learning, computer vision, and LLM applications. Completed a Machine Learning internship at FlyRank and built projects involving model training, evaluation, FastAPI model serving, RAG pipelines, and full-stack AI systems. Strongest areas include applied machine learning, computer vision, and LLM/RAG applications.

TECHNICAL SKILLS

Programming: Python, SQL, Java

Machine Learning: Scikit-learn, Classification, Regression, Clustering, Feature Engineering, Model Training, Model Inference, Model Evaluation, Cross-Validation, Hyperparameter Optimization, Error Analysis

Deep Learning & Computer Vision: TensorFlow, Keras, CNNs, Image Classification, Computer Vision, OpenCV, MediaPipe

LLM & NLP: RAG, Embeddings, Semantic Search, Vector Retrieval, Document Chunking, Prompt Engineering, LLaMA, Hugging Face

ML Engineering: FastAPI, REST APIs, Model Serving, ML Pipelines, Git, GitHub

Data: NumPy, Pandas, MySQL

EXPERIENCE

Machine Learning Intern — FlyRank

Remote

July 2026 – September 2026

- Applied data wrangling, embeddings, clustering, intent and opportunity modeling, and insight-to-action workflows during the FlyRank AI Internship.
- Delivered the “Google Search Ranking & Discoverability” capstone, reviewed and accepted by the lead track mentor.
- Built NLP data pipelines for search-data intelligence and collaborated with senior engineers on model evaluation and feature engineering.

SELECTED PROJECTS

MediSense AI — Healthcare ML Platform

Jan 2026 – July 2026

Python, TensorFlow, Keras, Scikit-learn, FastAPI, React, Groq LLaMA, Twilio

- Trained and deployed four ML/DL models for diabetes, heart disease, kidney disease, and chest X-ray pneumonia classification.
- Engineered end-to-end ML pipelines covering preprocessing, feature engineering, model training, evaluation, and FastAPI model serving.
- Integrated an LLM medical chatbot with multilingual support, voice input, WhatsApp alerts via Twilio, and a real-time analytics dashboard.
- Deployed on Vercel and PythonAnywhere; remediated exposed API credentials and resolved data-leakage and API-deprecation issues.

AI-Based Driver Safety System — Research Project

July 2026 – September 2026

OpenCV, MediaPipe, Computer Vision

- Developed a real-time computer vision pipeline for driver fatigue and unsafe-behavior detection from live video streams.
- Engineered a face-recognition module for driver authentication to help prevent unauthorized vehicle access.
- Developed the research methodology and evaluation approach for the system.

Bharatiya Kanoon AI — Multilingual Legal RAG

Jan 2026 – March 2026

Python, FastAPI, Embeddings, Vector Search, LLM, React

- Built a multilingual RAG pipeline for tenant-rights queries in English, Hindi, and Marwari-Rajasthani using the Rajasthan Rent Control Act 2001 and Model Tenancy Act 2021.
- Implemented document chunking, embeddings, semantic search, vector retrieval, and a FastAPI backend for grounded LLM responses.

EDUCATION

DBS Global University

Dehradun, India

B.Tech, Computer Science (AI/ML)

2024 – 2028

Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Design and Analysis of Algorithms, Software Engineering, DBMS, OOP

CERTIFICATIONS & ACHIEVEMENTS

- HackerRank Software Engineer Intern Certification — Credential ID: 042C07A26D05
- HackerRank 5-Star Python rating; completed Anthropic Academy coursework during the FlyRank AI Internship.
- Deployed MediSense AI as a live product across four disease-detection modules.